

CHAPTER#06"SOLIDS"

DADA OFFICIAL

DEFINITION:

"All those substances which possess definite shape as well as definite volume are known as Solids."

KINETIC THEORY OF SOLIDS

The assumptions of kinetic theory for solids are as follows:

- ✓ The particles in a solid are very closely packed due to strong attractive forces between the molecules.
- ✓ These molecules are present at a fixed position and are unable to move.
- ✓ They have definite shape because the particles are arranged in a fixed pattern.
- ✓ They possess only vibrational energy.

"BEHAVIOURS"

Following are the factors which affect the surface tension of liquids.

DIFFUSION:

Solids can diffuse but in very low extent. The diffusion in solids is due to the vibrational motion of the molecules.

COMPRESSION:

Solids molecules are much closed together and only slight unfilled spaces are felt, hence can be compressed to very little extent.

DEFORMATION:

Solids are deformed by high pressure because when some particles are dislocated, the force of attraction is so strong that the rearranged atoms are held equally well to their new neighbors.

MELTING:

When a solid is heated, the kinetic energy of the molecules increase and it overcomes the intermolecular attractions. As a result of this solid starts converting into liquid.

SUBLIMATION:

Some solids on heating directly change into vapors which are sublime solids and the phenomenon is known as sublimation.

In terms of KMT the intermolecular forces in such solids are very poor hence molecules of high energy at the surface of these solids overcome the attractive force and directly change into vapors.

"TYPES OF SOLIDS"

Solids are classified into two main classes:

1. Crystalline solids
2. Amorphous solids

"CRYSTALLINE SOLID"

DEFINITION:

"A solid in which particles are arranged in regular repeating three dimensional manners and have definite geometrical shape is known as crystalline solid."

EXAMPLES:

- ✓ Sugar
- ✓ Alum
- ✓ Diamond
- ✓ Metals etc.

"AMORPHOUS SOLID"

DEFINITION:

"Those solids in which particles are attached with each other with an indefinite arrangement and possess an indefinite geometrical shape are called amorphous solids."

EXAMPLES:

- ✓ Glass
- ✓ Plastic
- ✓ Rubber etc.

"DIFFERENCES"

CRYSTALLINE SOLIDS	AMORPHOUS SOLIDS
They have a definite geometrical shape	They do not have definite geometrical shape.
Particles are arranged in definite three dimensional pattern	Particles are not arranged in definite three dimensional pattern
They have sharp melting point	They do not have sharp melting point
They can be cleaved at fixed cleavage plane	They cannot be cleaved at fixed cleavage plane

They are anisotropic i.e. some of their physical properties are different in different directions.

They are symmetrical i.e. their appearance do not change by rotating them around an axis.

They are isotropic i.e. their physical properties are same in all directions

They are not symmetrical

TYPES OF CRYSTALS:

Based upon the bonding between the particles, the crystals are divided into four types.

"ATOMIC CRYSTALS"

DEFINITION:

"The crystals consist of atoms packed together by metallic bonding are called atomic crystal or metallic crystals."

EXPLANATION:

In metals the valence electrons are loosely held and move freely from one atom to another. They create a negative charge atmosphere around the positively charged metal ions. This negative charge atmosphere is called electron cloud.

PROPERTIES:

- ✓ They are good conductors of heat and electricity.
- ✓ They are lustrous.
- ✓ They possess high melting points.
- ✓ They are ductile and malleable.

"MOLECULAR CRYSTALS"

DEFINITION:

"The crystals consist of molecules that are held together by weak attractive forces are called molecular crystals."

EXPLANATION:

In these crystals following attractive forces are present.

✓ Hydrogen Bonding:

It is the electrostatic force of attraction between partially positive hydrogen of one molecule and the electronegative atom of other molecule.

✓ Vander Waal's Forces:

It is the attractive force developed between the atomic nuclei of one atom and electrons of other molecule.

PROPERTIES:

- ✓ They are soft.
- ✓ They have low melting point.
- ✓ They are usually non-conductors of heat and electricity.

"IONIC CRYSTALS"**DEFINITION:**

"The solids which consist of positively and negatively charged ions held together by strong electrostatic forces are called ionic crystals."

PROPERTIES:

- ✓ They are hard and brittle.
- ✓ They have high melting points.
- ✓ They conduct electricity in molten state or in the state of solution.
- ✓ Indefinite growth of crystals.

"COVALENT CRYSTALS"**DEFINITION:**

"In covalent crystals, atoms are held together by strong covalent bond and their breaking need high energy."

PROPERTIES:

- ✓ They are usually non-conductor of heat and electricity.
- ✓ They have high melting points.
- ✓ They have low densities.
- ✓ They have high value of the refractive index.

"METALLIC SOLID"**EXPLANATION:**

Particles of metallic crystals are metal atom according to electron sea theory, metal atoms use their all valence electrons in to the vacant spaces and becomes positively charged metal ions arranged themselves in a three-dimensional crystal lattice and bonded with sea of electrons. This unique force of attraction between metal cations and a sea of delocalized electrons is known as a metallic bond.

PROPERTIES:

- ✓ Metallic solids are identified by their malleability and ductility.

STATES OF MATTER-III: "SOLIDS"

Chapter #6

Metallic solids are looks like dull but when freshly cut, the inner layer appear shiny. This property is known as metallic luster.
The electrical conductivity of metallic solids is due to the free movement of delocalized electron in electric field.
Metals are good conductors of heat. Thus when a rod of metal is heated from one of its terminals, the free electrons of metal absorb heat, and hence their kinetic energy increases. These free electrons then tends to move towards the other end of the rod which is relatively cooler.

"ISOMORPHISM"

DEFINITION:

Substances having the same crystal structure are called Isomorphs and the phenomenon is known as Isomorphism."

EXAMPLES:

✓ CaCO_3 and NaNO_3
✓ ZnSO_4 and NiSO_4
✓ NaF and MgO

Trigonal crystals
Orthorhombic crystals
Cubic crystals

PROPERTIES:

- ✓ Their physical and chemical properties are different.
- ✓ They form mixed crystals in the solution.
- ✓ They have same empirical formula.
- ✓ They show the property of over growth.

"POLYMORPHISM"

DEFINITION:

The substance that exists in more than one crystalline form is called polymorphous and the phenomenon is known as Polymorphism."

EXAMPLES:

- ✓ Calcium carbonate exists in two different crystalline structures i.e.
 1. Calcite: It has trigonal structure.
 2. Aragonite: it has orthorhombic structure.
- ✓ Silver nitrate exists in two different crystalline structures i.e.
 1. Orthorhombic
 2. Rhombohedra.

PROPERTIES OF POLYMORPHOUS SUBSTANCE:

- ✓ They have same formula having same types of composition.

- ✓ They have same properties.
- ✓ Their crystalline systems are different.

"DIFFERENCE"

<u>ISOMORPHISM</u>	<u>POLYMORPHISM</u>
The substances having same crystal structure are called Isomorphous.	The substances that exist in more than one crystalline form are called Polymorphous.
They form mixed crystals.	They cannot form mixed crystals.
They exhibit property of overgrowth.	They do not exhibit the property of overgrowth.

"UNIT CELL"

DEFINITION:

"A basic structural unit which when repeated in a three dimensions generates a crystal is called Unit Cell."

OR

"The smallest volume of crystal which shows all the characters of its parent is called unit cell."

A unit cell has a definite geometry and hence it identifies on the basis of:

- ✓ Length of edges i.e. a , b and c
- ✓ Angle between edges i.e. α , β and γ

"CELL DIMENSIONS"

DEFINITION:

"The length and angles of edges of unit cell are known as Cell Dimensions or Cell Parameters."

"CRYSTAL LATTICE"

DEFINITION:

"If atoms, ions or molecules constituting a crystal are replaced by points and placed at the same places as in a unit cell, then the three dimensional array of points is called Crystal Lattice or Space Lattice."

"CRYSTAL SYSTEM"**DEFINITION:**

"Based on the difference in shape and dimension of unit cell of all crystals have been classified into seven groups called as crystal system."

S. No.	Crystal System	Axes	Angles	Example
1	Cubic system	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	NaCl, ZnS
2	Tetragonal system	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	SnO ₂
3	Orthorhombic system	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	KNO ₃
4	Trigonal / Rhombohedral	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$	AgNO ₃
5	Hexagonal system	$a = b \neq c$	$\alpha = \beta = 90^\circ \gamma = 120^\circ$	SiO ₂
6	Mono clinic system	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ \beta \neq 90^\circ$	CuSO ₄
7	Triclinic system	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	CuSO ₄ .5H ₂ O

"MELTING POINT"**DEFINITION:**

The temperature at which crystalline solids sharply changes into liquid are called Melting Point.

OR

The temperature at which there is a equilibrium between solid and liquid phase is called melting Point

FACTORS EFFECTING MELTING POINT:✓ **IMPURITY:**

Pure solid substances have a sharp melting point. The presence of impurity lowers the melting point of solids. Hence the melting point is used as criterion of the purity of solids.

✓ **PRESSURE:**

Those substances that expand on heating rise in their melting point when the pressure is increased.

"LATENT HEAT OF FUSION"**DEFINITION:**

"The heat energy required to change 1 gram solid into liquid at its melting point."

"CLOSE PACKING IN METALLIC SOLIDS"

It is believed that metal atoms are arranged most likely in the structure of a tightly packed ball which minimizes the empty space between them and justifies their high density.

There are three types of close packing found in metals.

➤ **HEXAGONAL CLOSE PACKING**

In this type of packing the empty spaces create due to three neighboring sphere of the first layer being filled by the alignment of spheres of second layer. The spheres of third layer are directly parallel to the sphere of the first layer so that an ABA-type arrangement is developed which covers 74% empty space.

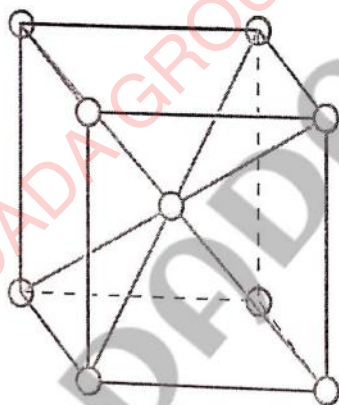
➤ **FACE CENTERED CUBIC CLOSE PACKING:**

In this type of packing the void spaces among the spheres of first layer is filled by spheres of the second layer, however, the spheres of third layer are displaced horizontally from first layer. This gives an ABCA arrangement that covers 74% of empty space.

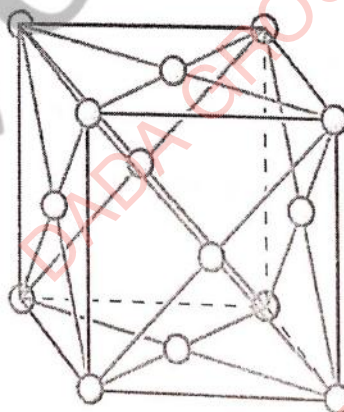
➤ **BODY-CENTERED CUBIC:**

This type of packing has a relatively big void space. The arrangement of the layer is ABAC type which covers up only 68% of empty space.

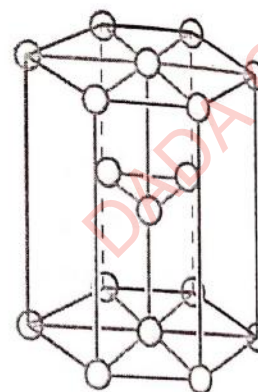
Common metallic crystal structures



body-centred cubic (bcc)



face-centred cubic (fcc)



hexagonal close-packed (hcp)